Predictive Model Plan – Geldium Delinquency Risk

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1. Model Logic (Generated with GenAI)

Chosen model. A class-weighted logistic regression that estimates each customer's probability of becoming delinquent (0–1). Customers are then ranked so the Collections team can prioritize outreach. A shallow decision tree (maximum depth 3) is trained alongside it as an explainable benchmark and to translate the scores into simple risk segments that front-line staff can act on.

Top 5 input features and how they contribute

Feature	How it is used	Expected effect on risk
1. Payment history	Missed_Payments (12 months) plus counts of Late and Missed months in Month_1–6	More missed or late payments → higher risk
2. Credit_Utilization	Capped at 100%, with a separate Over_Limit flag for the 4 accounts above their limit	Higher utilization → higher risk
3. Debt_to_Income_Ratio	Used as provided (decimal 0–1)	Higher ratio → higher risk
4. Credit_Score	Median-imputed for the 2 missing values	Lower score → higher risk
5. Account_Tenure	Years with Geldium	Shorter tenure → higher risk

Supporting inputs: Employment_Status (labels standardized), Income and Loan_Balance (median-imputed, each with a missing-value flag). Age and Location are deliberately **excluded** from the model as predictors and kept only for the fairness audit, because they can act as proxies for protected characteristics.

Workflow: from customer data to risk prediction

1. **Data ingestion and cleaning:** standardize Employment_Status labels, align utilization and debt-to-income units, and reconcile the 141 records where 12-month missed payments conflict with the 6-month history (findings from Task 1).
2. **Missing values:** median imputation for Income, Loan_Balance and Credit_Score, plus missing-value flags.
3. **Feature engineering:** payment-history counts, over-limit flag, and one-hot encoding of employment status; numerical features are standardized.
4. **Training:** stratified 5-fold cross-validation repeated 10 times; class weights offset the 16% delinquency rate, applied only to training folds.
5. **Output:** a probability score per customer, mapped to Low, Medium and High risk tiers using a threshold set with the Collections team based on their outreach capacity.
6. **Explanation:** model coefficients for the overall drivers and SHAP values for the top reasons behind each individual customer's score.

Simplified pseudocode (GenAI-assisted)

```
data  = load("Delinquency_prediction_dataset")
data  = standardize_labels(data); reconcile_payment_fields(data)
for col in [Income, Loan_Balance, Credit_Score]:
    data[col + "_Missing"] = is_missing(data[col])
    data[col] = fill_with_median(data[col])
X = [payment_history, utilization, over_limit, DTI, credit_score,
     tenure, employment, income, loan_balance, missing_flags]
y = Delinquent_Account                    # 1 = delinquent
model = LogisticRegression(class_weight="balanced")
scores = repeated_stratified_cv(model, X, y, folds=5, repeats=10,
```

```
            metrics=[AUC, PR_AUC, recall, precision, F1])
if scores meet deployment criteria (Section 3):
    risk_prob = model.predict_proba(new_customers)
    risk_tier = assign_tier(risk_prob, threshold=agreed_with_collections)
    explain   = shap_values(model, new_customers)  # top 3 reasons each
else:
    return to data remediation before any use in decisions
```

2. Justification for Model Choice

Logistic regression is the best fit for Geldium because the Collections team needs a ranked, explainable list of customers to contact rather than a black-box yes/no label. Its probability output maps directly onto outreach priority, and its coefficients show exactly how each factor raises or lowers risk. That transparency is what credit regulators and fair-lending rules expect when a customer is treated differently because of a model, and it also makes the model easy for Geldium's risk team to audit and challenge. The model also fits the data: with 500 customers and only 80 delinquent cases, more complex models such as neural networks or deep tree ensembles would memorize noise rather than learn stable patterns. Logistic regression is fast to train, cheap to run on every customer daily, and simple to monitor, so drifting coefficients or scores are easy to spot. The trade-off is that it captures mainly linear effects and can miss interactions (for example, high utilization mattering more for short-tenure customers). The shallow decision tree benchmark is included to detect such interactions, and if it consistently outperforms the regression, its rules can be added as features.

Option	Strengths	Trade-offs	Decision
Logistic regression	Probability scores; transparent coefficients; stable on small data; easy to deploy and monitor	Mostly linear; interactions must be engineered	Primary model
Decision tree (depth 3)	Clear if-then segments; captures interactions; handles mixed data	Unstable on small samples; coarse probabilities	Benchmark and segmentation
Neural network	Captures complex patterns on large data	Black box; overfits 500 records; hard to justify to regulators	Rejected

3. Evaluation Strategy

Metrics and how to interpret them

Metric	What it tells Geldium	Target before deployment
AUC-ROC	How well the model ranks risky customers above safe ones; 0.5 is random guessing	≥ 0.70
Recall	Share of truly delinquent customers the model catches; missed cases mean lost money	≥ 0.70 at the agreed threshold
Precision	Share of flagged customers who really are delinquent; low precision wastes outreach effort	≥ 2× the base rate (≥ 0.32)
F1 score and PR-AUC	Balance of recall and precision; more informative than AUC-ROC when positives are rare	Clearly above the 0.16 base rate
Accuracy	Reported for completeness only: a model that predicts "not delinquent" for everyone already scores 84%	Not used for decisions

Metric	What it tells Geldium	Target before deployment
Confusion matrix	Shows the actual number of missed delinquents vs false alarms at the chosen threshold	Reviewed with Collections

Reliability checks

- Repeated stratified cross-validation, reporting the average and spread of each metric so results do not depend on one lucky split.
- Threshold tuning against Collections capacity (how many customers the team can contact per cycle), rather than defaulting to 0.5.
- Calibration check so that a predicted 30% risk corresponds to roughly 30% of such customers actually becoming delinquent.
- After launch: monthly monitoring of score distribution and recall, with retraining when performance drops below target.

Baseline result and go/no-go decision. As a proof of concept, this pipeline was run on the current dataset. The logistic regression reached a cross-validated AUC of 0.47 (±0.05), with precision of 0.14 against a 0.16 base rate, and the decision tree reached 0.49. Both are at the level of random guessing, which confirms the Task 1 finding that the current fields carry almost no usable signal. The model therefore **does not meet the deployment criteria** and should not be used to make customer decisions until the data issues are resolved and the targets above are met. Model improvement steps, in order: fix field definitions and conflicts, add stronger behavioral features (such as recency of the last missed payment and changes in utilization over time), then re-test.

Bias detection and mitigation

- **Disparate impact:** compare the share of customers flagged High risk across age bands and locations; a ratio below 0.80 between any group and the most favorably treated group (the four-fifths rule) triggers review.
- **Equalized odds:** compare recall and false-positive rates across the same groups; a gap above 10 percentage points triggers review.
- **Proxy check:** test whether retained features (for example, Credit_Card_Type "Student") are strongly linked to age, and remove or adjust them if so.
- **Mitigation options:** reweighting training data, group-specific threshold review, or dropping proxy features, with every change documented and re-tested.

Ethical considerations

- The model supports, not replaces, human judgment: a collections officer reviews High-risk cases before any action that affects a customer.
- Risk scores should trigger supportive interventions (payment reminders, hardship plans) rather than punitive ones such as automatic penalty rates.
- Each customer's score must be explainable in plain language, with a route for customers to question decisions and correct inaccurate data.
- GenAI was used to draft the model logic and evaluation plan; every recommendation and number here was checked against the data and established methods, since GenAI explanations can be inaccurate.